

■ 2.6.7 Advanced Topic: Mimicking Mathematical Notation

In creating your own output forms, the goal is often to mimic standard mathematical notation. One of the most common elements of standard notation is the use of subscripts and superscripts. Output forms for these are built in to *Mathematica*.

<code>Subscripted[f[x₁, x₂, ...]]</code>	a function with arguments given as subscripts
<code>Subscripted[f[args], n]</code>	a function with the first <i>n</i> arguments as subscripts
<code>Subscripted[f[args], {d₀, d₁}, {u₀, u₁}]</code>	arguments <i>d</i> ₀ through <i>d</i> ₁ as subscripts; <i>u</i> ₀ through <i>u</i> ₁ as superscripts

Output forms that give subscripts and superscripts.

This gives an output form with both arguments of *a* as subscripts.

```
In[1]:= Subscripted[a[1, 2]]
Out[1]= a1,2
```

This defines *a* with any number of arguments to be output with the arguments as subscripts.

```
In[2]:= Format[a[x_]] := Subscripted[a[x]]
```

The subscripts can themselves contain superscripts.

```
In[3]:= a[x^2, y^2]
Out[3]= ax2, y2
```

You can specify what sequences of arguments to a function should be subscripts or superscripts. The notation for each sequence is the same as in **Take** and **Drop**. Any arguments that you do not explicitly specify as subscripts or superscripts are output in standard functional notation.

This outputs the first two arguments as subscripts, and the remainder in standard functional notation.

```
In[4]:= Subscripted[f[a, b, c, d], 2]
Out[4]= fa,b[c,d]
```

This gives the last argument as a subscript.

```
In[5]:= Subscripted[f[a, b, c, d], -1]
Out[5]= fd[a,b,c]
```

Here the first two arguments of *f* are printed as subscripts, and the last one is a superscript. The remaining argument is given in standard functional notation.

```
In[6]:= Subscripted[f[a, b, c, d], 2, -1]
Out[6]= fa,bd[c]
```

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Mathematica allows you to go beyond just having subscripts and superscripts, and to define any kind of two-dimensional output form. In a typical case, you can build up a complicated two-dimensional form from a combination of horizontal (`SequenceForm`) and vertical (`ColumnForm`) “boxes”.

<code>SequenceForm</code> [f_1 , f_2 , ...]	a sequence of objects, arranged horizontally
<code>ColumnForm</code> [{ f_1 , f_2 , ...}]	a column of objects
<code>ColumnForm</code> [<i>list</i> , <i>h</i> , <i>v</i>]	a column with horizontal alignment <i>h</i> (Left, Center or Right), and vertical alignment <i>v</i> (Below, Center or Above)
<code>Subscript</code> [<i>x</i>]	a subscript
<code>Superscript</code> [<i>x</i>]	a superscript

Objects for building up two-dimensional output forms.

`SequenceForm` concatenates the output forms of expressions, with no spaces in between.

`In[7]:= SequenceForm["-+-", 1/x, "-+-"]`

`Out[7]=`
$$-+-\frac{1}{x}+-$$

`ColumnForm` arranges lists of expressions in vertical columns.

`In[8]:= ColumnForm[{"-+-", x + y, "-+-"}]`

`Out[8]=`
$$\begin{array}{c} -+- \\ x + y \\ -+- \end{array}$$

You can build up more complicated two-dimensional structures by nesting `SequenceForm` and `ColumnForm`.

`In[9]:= SequenceForm[%, "****", %]`

`Out[9]=`
$$\begin{array}{cc} -+- & ****-+- \\ x + y & x + y \\ -+- & -+- \end{array}$$

`Subscript` and `Superscript` are output as subscripts and superscripts.

`In[10]:= f[Subscript[x], y, Superscript[z^2]]`

`Out[10]=`
$$f\left[\substack{x}, y, \supset z^2\right]$$